

SAT UZBEKISTAN

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Module-1

1.

A right square pyramid has a height of 11 centimeters (cm). A second right square pyramid has a height of 22 centimeters. The area of the base of each of the two pyramids is 100 cm^2 . Which of the following is closest to the difference in the surface area of the second pyramid and the surface area of the first pyramid, in cm^2 ?

- A) 209
- B) 220
- C) 551
- D) 367

2.

A right square pyramid has a total surface area of 36,864 square inches, and the combined surface area of the four lateral faces of this pyramid is 20,480 square inches. What is the height, in inches, of this pyramid?

- A) 48
- B) 64
- C) 80
- D) 128

3.

A polygon has 16 sides has a perimeter of 126. The length of one side is 6 inches. The other 15 sides have equal lengths. What is the length, in inches, of one of the 15 sides with equal lengths?



4.

	Volume (cubic units)
Right circular cylinder A	112π
Right circular cylinder B	$3,024\pi$

The table shows the volume of two similar solids, right circular cylinder A and right circular cylinder B. The radius of right circular cylinder A is 4 units. The surface area of right circular cylinder A is $k\pi$ square units, and the surface area of right circular cylinder B is $n\pi$ square units, where k and n are constants. What is the value of $n - k$? (The surface area of a right circular cylinder with radius r and height h is $2\pi r^2 + 2\pi rh$.)

5.

$$c(r) = \frac{2}{5}(2\pi r)$$

The function c gives the partial length, in centimeters (cm), of the circumference of a circle with a radius that measures r cm. According to the function, for each increase 18 cm in the length of radius, this partial length of circumference increases by $k\pi$ cm. What is the value of k ?

6.

Point F lies on a unit circle in the xy -plane and has coordinates $(1,0)$. Point G is the center of the circle and has coordinates $(0,0)$. Point H also lies on the circle and has coordinates $(-1,y)$, where y is a constant. Which of the following could be the positive measure of angle FGH , in radians?

- A) $\frac{27\pi}{2}$

B) $\frac{27\pi}{2}$
- C) 22π

D) 23π

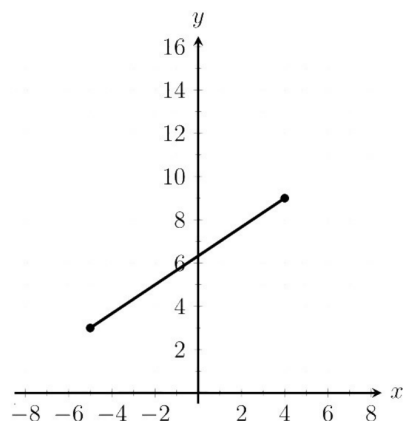
7.

$$x^2 - 10x + y^2 - 6y - 47 = 0$$

In the xy -plane, the graph of the given equation is a circle. If this circle is inscribed in a square, what is the perimeter of the square?

- A) 18 C) 72
B) 36 D) 188

8.



The line segment shown in the xy -plane represents one of the legs of a right triangle. The area of this triangle is $48\sqrt{13}$ square units. What is the length, in units, of the other leg of this triangle?

- A) 16 C) $3\sqrt{13}$
B) 32 D) 24

9.

In triangle FGH and triangle KLM , the measures of angles G and L are each 60° . The lengths of GH and LM are each 34 centimeters, and $\frac{GH}{GF} = \frac{LM}{LK}$. Which additional piece of information would be necessary to prove that triangle FGH is congruent to triangle KLM ?

- A) The lengths of FG and KM are each 17 centimeters.
B) The measures of angles H and M are each 30° .
C) The measures of angles F and K are each 90° .
D) No additional information is necessary.

10.

A triangle inequality theorem states that the sum of any two sides of a triangle must be greater than the length of the third side. If a triangle has side length of 8 and 13, which of inequality represents the possible length, x , of the third side of the triangle?

- A) $x < 21$
- B) $x > 21$
- C) $5 < x < 21$
- D) $x < 5$ or $x > 21$

11.

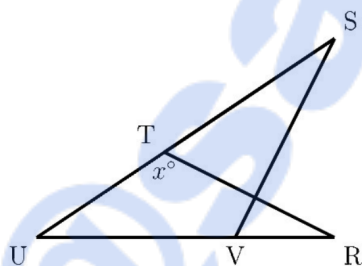
Triangle QRS is similar to triangle FGH such that Q, R , and S correspond to F, G , and H , respectively. Each side of triangle FGH has $\frac{1}{4}$ the length of its corresponding side in triangle QRS . In triangle FGH , the measure of angle F is 35° , the measure of angle G is 8° , and $FG = a$. In triangle QRS , the measure of angle Q is b° and $QR = 20$. What is the value of $a + b$?

12.

In isosceles triangle ABC , sides AB and AC are congruent. Point D divides BC such that the length of $\overline{BD}$ is $\frac{5}{7}$ of the length of $\overline{BC}$. Point E lies on side AB and point F lies on side AC such that when segments DE and DF are drawn, angle BED is congruent to angle CFD . If the length of $\overline{BE}$ is 10, what is the length of $\overline{CF}$?

- | | |
|-------|-------|
| A) 4 | C) 50 |
| B) 14 | D) 70 |

13.



Note: Figure not drawn to scale

In the figure, $RT = TU$, the measure of angle VST is 23° , and the measure of angle RVS is 34° . What is the value of x ?

14.

Triangle ABC is similar to triangle DEF , where angle A corresponds to angle D and angles C and F are right angles. The length $\overline{AB}$ is 2.4 times the length of $\overline{DE}$. If $\tan A = \frac{21}{20}$, what is the value of $\sin D$?

15.

In triangle PQR , the measures of angle P is $(3x+5)^\circ$, the measure of angle Q is $(2x+9)^\circ$, and the measure of angle R is $(4y+5)^\circ$. If side QR is extended through point R to point S , and the measure of angle PRS is $(x+y)^\circ$, what is the value of $x+y$?

16.

A researcher is designing a study to investigate the average number of hours a student at a high school spends reading per day. The researcher will report the average number of hours a student at the high school spends reading with an associated margin of error. The researcher is considering using a random sample of either 105 or 210 students from the high school. Which of the following could be the most likely effect of using the larger random sample compared to the smaller random sample?

- A) The reported margin of error would be lower.
- B) The reported margin of error would be higher.
- C) The reported average number of hours would be lower.
- D) The reported average number of hours would be higher.

17.

Value	Data set <i>A</i> frequency	Data set <i>B</i> frequency
40	2	8
43	4	7
46	5	5
49	7	4
52	8	2

Data set *A* and data set *B* each consist of 26 values. The table shows the frequencies of the values for each data set. Which of the following statements best compares the means of the two data sets?

- A) The mean of data set *A* is greater than the mean of data set *B*.
- B) The mean of data set *A* is less than the mean of data set *B*.
- C) The mean of data set *A* is equal to the mean of data set *B*.
- D) There is not enough information to compare the means of the data sets.

18.

An environmental scientist is investigating the volatility of 175 organic compounds by finding the boiling point of each compound. The mean boiling point of all 175 compounds is 167 degrees Celsius. The scientist classifies each of these compounds as either semivolatile or volatile. Of the 175 compounds, 50 compounds were classified as semivolatile, and these 50 compounds have a mean boiling point of 327 degrees Celsius. The remaining 125 compounds were classified as volatile. What is the mean boiling point, in degrees Celsius, of the 125 compounds classified as volatile compounds?

19.

	Even integers	Odd integers	Total
List A	12	44	56
List B	40	4	44
Total	52	48	100

The given table shows the number of even and odd integers contained in list A and list B. No integer is contained in both lists. What fraction of even integers represented in the table are in list A?

- A) $\frac{12}{52}$
- B) $\frac{12}{56}$
- C) $\frac{52}{100}$
- D) $\frac{52}{56}$

20.

For 100 neurons, the table summarizes the distribution of classification and cell body diameter:

Classification	Cell body diameter (micrometers)		
	Less than 20	20 to 30	Greater than 30
Sensory neuron	13	7	3
Motor neuron	0	15	18
Interneuron	10	34	0

One of these neurons will be selected at random. What is the probability of selecting a neuron with a cell body diameter that is less than or equal to 30 micrometers, given that it is not classified as a motor neuron?

21.

The table summarizes the distribution of age and assigned group for 105 participants in a study.

	0-9 years	10-19 years	20+ years	Total
Group A	5	19	11	35
Group B	4	8	23	35
Group C	26	8	1	35
Total	35	35	35	105

One of these participants will be selected at random. What is the probability of selecting a participant from group A, given that the participant is at least 10 years of age

22.

The speeds of particles A, B, and C are a meters per second, b meters per second, and c meters per second, respectively. If the speed of particle A is 6,000% of the speed of particle C and the speed of particle C is 0.008% of the speed of particle B, which expression represents the value of $a + b$ in terms of c ?

A) $12,560c$

C) $6,008c$

B) $6,125c$

D) $1,850c$

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Module-2

1.

At the start of a stock trading competition, John's and Julia's stock portfolios were of equal value. At the end of the stock trading competition, John's portfolio had increased by 330%. Julia's portfolio has increased by 2,480%. At the end of the competition Julia's portfolio is worth $p\%$ more than John's portfolio, what is the value of p ?

2.

For the exponential function f , the value of $f(3)$ is k , where k is a constant. Which of the following equivalent forms of the function f shows the value of k as the coefficient or the base?

- A) $f(x) = 3(4)(4)^{x+1}$
- B) $f(x) = 3,072(4)(4)^{x-4}$
- C) $f(x) = 12(4)(4)^x$
- D) $f(x) = 768(4)(4)^{x-3}$

3.

In the given pair of equations a and b are constants. The graph of this pair of equations in the xy -plane is a pair of perpendicular lines. Which of the following pairs of equations also represents a pair of perpendicular lines?

$$\begin{aligned} 13x + 17y &= 10 \\ ax + by &= 10 \end{aligned}$$

- A) $39x + 17y = 10$
 $ax - 3by = 10$
- B) $52x + 34y = 10$
 $2ax + 4by = 10$
- C) $65x + 17y = 10$
 $5ax + by = 10$
- D) $13x - 17y = 10$
 $ax + by = 10$

4.

The given quadratic function $54x^4 + 219x^2 + 105$ has factors in the form $(k)(ax^2 + b)(cx^2 + d)$. If a, b, c, d , and k are integers, what is the smallest possible value of ab ?

5.

In triangle ABC , angle C is a right angle, point D lies on $\overline{AB}$, point E lies on $\overline{BC}$, and $\overline{DE}$ is parallel to $\overline{AC}$. If the length of $\overline{AB}$ is 39 units, $\overline{DE}$ is 8 units, the length of $\overline{AC}$ is greater than the length of $\overline{BC}$, and the area of triangle ABC is 270 units, what is the length of $\overline{BE}$, in units?

6.

A rectangular gameboard is divided into 480 square regions of equal area k , in square units. The length of the gameboard is 7.5 times the width of the gameboard. If the width of the gameboard is $x\sqrt{k}$ units, what is the value of x ?

7.

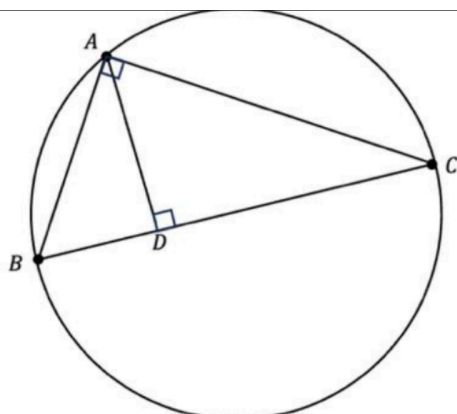
The function f is defined by $f(x) = ax^2 + bx + c$, where a, b and c are constants. The graph of $y = f(x)$ in the xy -plane passes through the points $(13, 0)$ and $(-6, 0)$. Which of the following is the value of $a + b$ in terms of a ?

- A) $7a$
- B) $6a$
- C) $-6a$
- D) $-7a$

8.

The edge length of the square base and the height of a right square pyramid are integers, and the pyramid has a volume of 16,464 cubic inches. If the slant height is 35 inches, what is the height, in inches, of this pyramid?

9.



Note: Figure not drawn to scale.

In the figure shown, points A , B , and C lie on the circle and $AC > AB$ and $AD = \sqrt{592}$. The diameter of the circle is 82. If $\frac{CD}{BD} = k$, what is the value of k ?

10.

For the linear function k , $k(a) = -24$, where a is a constant, $k(4) = 30$, and the slope of the graph $y = k(x)$ in the xy -plane is 7. For the linear function p , $p(a) = 26$ and $p(6) = 38$. What is the slope of the graph of $y = p(x)$ in the xy -plane?

11.

$$r(t) = 55t - 2t^2$$

The function r is defined by the given equation. The function s is defined by $s(t) = r(t) + 1$. Which expression represents the maximum value of $s(t)$?

- A) $1 + \left(\frac{55}{2}\right)^2$ C) $1 - 2\left(\frac{55}{4}\right)^2$
 B) $1 + 2\left(\frac{55}{4}\right)^2$ D) $1 - \left(\frac{55}{2}\right)^2$

12.

$$x^2 + 11x + 19 = 0$$

One solution to the given equation can be written as $x = \frac{-11-\sqrt{k}}{2}$, where k is a constant. What is the value of k ?

13.

The function f is defined by $f(x) = ax^2 + bx + c$, where a , b , and c are constants and $1 < a < 4$. The graph of $y = f(x)$ in the xy -plane passes through the points $(11, 0)$, and $(-5, 0)$. If a is an integer, what could be the value of $a + b$?

14.

$$f(x) = (x + a)(x + b)$$

For the given function $f(x)$, where a and b are integer constants, $f(-21) > 0$, $f(-15) > 0$, and $f(-18) < 0$, what is one possible value of $a + b$?

15.

In a set of three consecutive odd integers, where the integers are ordered from least to greatest, the first integer is represented by k . The product of 17 and the second odd integer in the set is at most the value of 40 less than the sum of the first and third odd integers in the set. What is the greatest possible value of k ?

16.

$$g(x) = 8(x - d)(x - e)(x - f)$$

The function g is defined by the given equation, where d , e , and f are distinct constants. When $d < x < f$, the value of $g(x)$ is positive. The graph of $y = g(x)$ in the xy -plane contains the point (p, q) where p and q are constants. If $q = 7$, which of the following could be true?

- I. $e < p$
- II. $p < d$
- III. $f < p < e$

- A) I only
- B) II only
- C) I and II only
- D) I and III only

17.

For the given parabola $ax^2 + bx + c$, a , b , and c are constants. The x -intercepts of the parabola are at $(14, 0)$ and $(k, 0)$. If $f(27) = f(11)$, what is the value of k ?

18.

$$f(x) = x^2 + 5x - 6$$

$$g(x) = x^2 + 2x - 35$$

The quadratic function $f(x)$ has 2 solutions j and k , where $j < k$. The quadratic function $g(x)$ has 2 solutions l and m , where $l < m$. The quadratic function $h(x) = x^2 + 7x + c$ has solutions of $k + l$ and $m + j$, and can be rewritten as $(x + a)(x + b)$, what is the value of c ?

19.

$$2x^2 - 16x - 19 = 0$$

One solution to the given equation can be written as $x = a - \frac{1}{2}\sqrt{b}$, where a and b are integer constants. What is the value of $a + b$?

20.

The function f is defined by $f(x) = 3^x$. The function g is an increasing linear function. In the xy -plane, the graphs of $y = f(x)$ and $y = g(x)$ intersect at two points (a, j) and (b, k) , where $j < k$. When $g(x) > f(x)$, which of the following must be true?

- A) $x > k$
- B) $x < j$
- C) $x < j$ or $x > b$
- D) $a < x < b$

21.

The expression $6x^4 + 35x^2 + 11$ can be rewritten as $(3x^2 + j)(2x^2 + k)$, where j and k are positive integers, or as $(3x^2 + l)(2x^2 + m)$, where l and m are positive nonintegers, what is the value of $k + l$?

22.

There are red, blue, green, and black marbles in a bag. There are 26 black marbles, 8 green marbles, and 26 blue and red marbles. Given that a randomly selected marble is not black, the probability of selecting a red marble is $\frac{5}{17}$. What is the number of blue marbles in the bag?